

# Analysis of Variance (ANOVA)

Rex Shao-Yu Wang

Department of Agronomy, National Taiwan University

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# Outline

Workflow

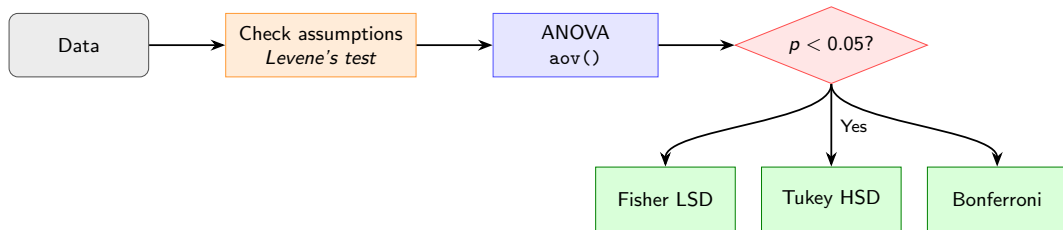
ANOVA

Multiple Comparison

Assumptions Check

Lab Report

# ANOVA Workflow



# ANOVA

A statistical method used to determine whether there are significant differences between the means of three or more independent groups.

## Hypothesis:

$H_0 : \mu_1 = \mu_2 = \dots = \mu_k$  (All group means are equal)

$H_1$  : At least one group mean is different from the others

## Test Statistic:

$$F_0 = \frac{MS_A}{MS_W} = \frac{SS_A/(k-1)}{SS_W/(N-k)} \sim F_{k-1, N-k}$$

where  $k$  = number of groups,  $N$  = total sample size.

## Idea:

If among-group variation  $\gg$  within-group variation, then  $F_0$  is large  $\Rightarrow$  evidence against  $H_0$ .

# ANOVA

## ANOVA Table:

| Source          | df      | SS     | MS                      | $F_0$         |
|-----------------|---------|--------|-------------------------|---------------|
| Among (Between) | $k - 1$ | $SS_A$ | $MS_A = SS_A / (k - 1)$ | $MS_A / MS_W$ |
| Within (Error)  | $N - k$ | $SS_W$ | $MS_W = SS_W / (N - k)$ |               |
| Total           | $N - 1$ | $SS_T$ |                         |               |

$$\sum_{i=1}^k \sum_{j=1}^n (y_{ij} - \bar{y}_{..})^2 = \sum_{i=1}^k \sum_{j=1}^n (\bar{y}_{i.} - \bar{y}_{..})^2 + \sum_{i=1}^k \sum_{j=1}^n (y_{ij} - \bar{y}_{i.})^2$$

$$SS_T = SS_A + SS_W$$

Total SS                      Among-group SS                      Within-group SS  
                                     Between-group SS                      Residual SS  
                                     Treatment SS                      Error SS

## Rejection Region:

Reject  $H_0$  if  $F_0 > F_{\alpha, k-1, N-k}$  (equivalently,  $p < \alpha$ ).

## Example: Plum weights

A farmer wants to compare the weights (in grams) of **4 plum varieties**. Five plums were randomly selected from each variety ( $N = 20$ ).

| Variety (5 plums each) | Weights (g)        |
|------------------------|--------------------|
| Red (Crimson Globe)    | 62, 60, 71, 55, 48 |
| Purple (Black Amber)   | 63, 57, 52, 41, 43 |
| Green (Greengage)      | 42, 50, 41, 37, 45 |
| Yellow (Mirabelle)     | 32, 39, 51, 30, 35 |

### Research question:

Do the four plum varieties have significantly different mean weights?

### Hypotheses:

$$H_0 : \mu_R = \mu_P = \mu_G = \mu_Y$$

$$H_1 : \text{at least one differs}$$

# ANOVA in R

## Code

```
model <- aov(weight ~ plum, data = data_plum)
summary(model)
```

## Output

|           | Df | Sum Sq | Mean Sq | F value | Pr(>F)     |
|-----------|----|--------|---------|---------|------------|
| plum      | 3  | 1363.4 | 454.5   | 7.222   | 0.00279 ** |
| Residuals | 16 | 1006.8 | 62.9    |         |            |

## Conclusion

$F_0 = 7.22$ ,  $p < 0.05 \Rightarrow$  Reject  $H_0$ .

Mean weights differ significantly among plum varieties.

# Multiple Comparisons

## Purpose:

ANOVA tells us *whether* group means differ, but **not which ones**.

Post-hoc tests identify **which pairs of groups** are significantly different.

## Three methods we will cover:

- ▶ **Fisher's LSD** (Least Significant Difference)
- ▶ **Tukey's HSD** (Honestly Significant Difference)
- ▶ **Bonferroni correction**

## Fisher's LSD

A pairwise comparison method that performs  $t$ -tests between all pairs of group means **without adjusting** the significance level.

**Hypothesis (for each pair  $i, j$ ):**

$$H_0 : \mu_i = \mu_j$$

$$H_1 : \mu_i \neq \mu_j$$

**Test Statistic:**

$$LSD = t_{\alpha/2, N-k} \cdot \sqrt{MS_W \left( \frac{1}{n_i} + \frac{1}{n_j} \right)}$$

where  $MS_W$  is the within-group mean square from the ANOVA.

**Decision Rule:**

Reject  $H_0$  if  $|\bar{y}_i - \bar{y}_j| > LSD$  (the two means differ significantly).

# Fisher's LSD in R

## Run pairwise comparisons with `LSD.test()`

```
library(agricolae)
LSD.test(model, "plum", console = TRUE)
```

## Output (selected):

```
$groups
      weight groups
Red       59.2     a
Purple   51.2    ab
Green    43.0    bc
Yellow   37.4     c
```

Groups sharing the same letter are **not** significantly different.

⇒ Red differs from Green and Yellow; Purple differs from Yellow.

# Tukey's HSD

A pairwise comparison method that uses the **studentized range distribution** to control the family-wise error rate.

**Hypothesis (for each pair  $i, j$ ):**

$$H_0 : \mu_i = \mu_j$$

$$H_1 : \mu_i \neq \mu_j$$

**Test Statistic:**

$$HSD = q_{\alpha, k, N-k} \cdot \sqrt{\frac{MS_W}{n}}$$

where  $q$  is the critical value from the studentized range distribution,  $k$  = number of groups, and  $n$  = sample size per group.

**Decision Rule:**

Reject  $H_0$  if  $|\bar{y}_i - \bar{y}_j| > HSD$  (the two means differ significantly).

# Tukey's HSD in R

Run Tukey's pairwise comparisons with `TukeyHSD()`

```
TukeyHSD(model)
```

Output:

```
$plum
      diff      lwr      upr      p adj
Purple-Red    -8.0  -22.353    6.353  0.4172
Green-Red   -16.2  -30.553   -1.847  0.0238
Yellow-Red  -21.8  -36.153   -7.447  0.0023
Green-Purple  -8.2  -22.553    6.153  0.3998
Yellow-Purple -13.8  -28.153    0.553  0.0617
Yellow-Green  -5.6  -19.953    8.753  0.6943
```

Significant pairs ( $p < 0.05$ ): **Green-Red**, **Yellow-Red**.

⇒ Red plums are significantly heavier than Green and Yellow.

## Bonferroni Correction

A simple adjustment method that divides the significance level  $\alpha$  by the number of comparisons to control the family-wise error rate.

**Hypothesis (for each pair  $i, j$ ):**

$$H_0 : \mu_i = \mu_j$$

$$H_1 : \mu_i \neq \mu_j$$

**Adjusted Significance Level:**

$$\alpha' = \frac{\alpha}{L}, \quad \text{where } L = \binom{k}{2} = \frac{k(k-1)}{2}$$

For  $k = 4$  groups:  $L = 6$ , so  $\alpha' = 0.05/6 \approx 0.0083$ .

**Decision Rule:**

Perform pairwise  $t$ -tests, and reject  $H_0$  only if  $p < \alpha'$ .

# Bonferroni Correction in R

## Run pairwise $t$ -tests with Bonferroni adjustment

```
pairwise.t.test(data_plum$weight, data_plum$plum,  
                p.adjust.method = "bonferroni")
```

## Output:

|        | Red    | Purple | Green  |
|--------|--------|--------|--------|
| Purple | 0.6964 | -      | -      |
| Green  | 0.0316 | 0.6275 | -      |
| Yellow | 0.0029 | 0.0935 | 1.0000 |

Significant pairs ( $p < 0.05$ ): **Red–Green, Red–Yellow.**

⇒ Same significant pairs as Tukey, but with larger  $p$ -values.

# ANOVA Assumptions

Before applying ANOVA, the following assumptions should be checked:

- ▶ **Independence**

Observations within and between groups are independent of each other.  
*Ensured by proper experimental design and random sampling.*

- ▶ **Normality**

Residuals should be approximately normally distributed.  
*Robust to mild violations under the Central Limit Theorem.*

- ▶ **Homogeneity of Variances**

All groups have equal variances ( $\sigma_1^2 = \sigma_2^2 = \dots = \sigma_k^2$ ).  
*Checked by Levene's test.*

**Note:** If assumptions are violated, results may be unreliable. Data transformations (e.g., log,  $\sqrt{\cdot}$ ) or non-parametric alternatives may be considered.

# Homogeneity of Variances

## Concept:

ANOVA assumes that all groups have **equal variances**:

$$\sigma_1^2 = \sigma_2^2 = \cdots = \sigma_k^2$$

Violation of this assumption can inflate the Type I error rate, especially with unbalanced designs.

**Levene's test** — a formal hypothesis test for equal variances:

- ▶  $H_0: \sigma_1^2 = \sigma_2^2 = \cdots = \sigma_k^2$  (variances are equal)
- ▶  $H_1$ : at least one variance differs
- ▶ Reject  $H_0$  if  $p < 0.05$  (i.e., homogeneity is violated)

## Idea:

Levene's test performs an ANOVA on the **absolute deviations** from each group's mean (or median). If groups have similar spread, the deviations should not differ across groups.

# Homogeneity Check in R

## Run Levene's test with `leveneTest()`

```
library(car)  
leveneTest(weight ~ plum, data = data_plum)
```

## Output:

```
Levene's Test for Homogeneity of Variance (center = median)  
      Df F value Pr(>F)  
group  3   0.5121 0.6797  
      16
```

$p = 0.68 > 0.05 \Rightarrow$  **Homogeneity assumption is satisfied.**

# Lab Report

- ▶ The report file will be in the NTU COOL Assignment section
- ▶ Submit your file in pdf format
- ▶ Deadline: 5/24 (Sun) 23:59
- ▶ Late submission is not accepted